Лабораторная работа №15

Динамическая маршрутизация

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Раздел 1

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Цель работы

Настроить динамическую маршрутизацию между территориями организации.

Выполнение работы

Для начала настроим OSPF на маршрутизаторе `msk-donskaya-agko-gw-1`.
Включение OSPF на маршрутизаторе предполагает, во-первых, включение процесса OSPF командой `router ospf`, во-вторых — назначение областей (зон) интерфейсам с помощью команды `network area` (рис. #fig:002).

```
msk-donskaya-agko-gw-1>enable
Password:
msk-donskaya-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-agko-gw-1(config)#router ospf 1
msk-donskaya-agko-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-agko-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-agko-gw-1(config-router)#^Z
msk-donskaya-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
```

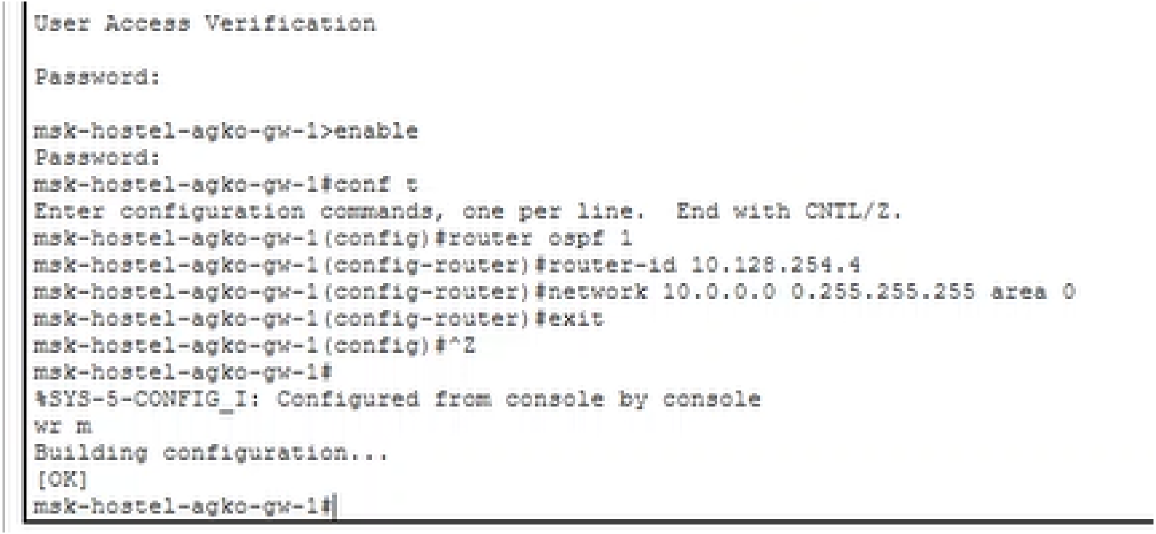
Рисунок 1: Настройка OSPF на маршрутизаторе msk-donskaya-agko-gw-1 (включение процесса OSPF, назначение областей интерфейсам)

Проверим состояние протокола OSPF на маршрутизаторе `msk-donskaya-agko-gw-1`. Маршрутизаторы с общим сегментом являются соседями в этом сегменте. Соседи выбираются с помощью протокола Hello. Команда `show ip ospf neighbor` показывает статус всех соседей в заданном сегменте. Команда `show ip route` выводит информацию из таблицы маршрутизации (рис. #fig:003):

```
[OK]
msk-donskaya-agko-gw-1#sh ip ospf
 Routing Process "ospf 1" with ID 10.128.254.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 8
        Area has no authentication
        SPF algorithm executed 1 times
        Area ranges are
        Number of LSA 1. Checksum Sum 0x00312a
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

msk-donskaya-agko-gw-1#sh ip ospf neighbor

msk-donskaya-agko-gw-1#sh ip ospf route
                                  ^
% Invalid input detected at '^' marker.

msk-donskaya-agko-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0
```

Далее приступим к настройке маршрутизатора msk-q42-agko-gw-1, маршрутизирующего коммутатора msk-hostel-agko-gw-1 и маршрутизатора sch-sochi-agko-gw-1 (рис. #fig:004 – #fig:006):

```
Password:

msk-q42-agko-gw-1>enable
Password:
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#router ospf 1
msk-q42-agko-gw-1(config-router)#router-id 10.128.254.2
msk-q42-agko-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-q42-agko-gw-1(config-router)#exit
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-q42-agko-gw-1#
```



```
User Access Verification

Password:

msk-hostel-agko-gw-1>enable
Password:
msk-hostel-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-agko-gw-1(config)#router ospf 1
msk-hostel-agko-gw-1(config-router)#router-id 10.128.254.4
msk-hostel-agko-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-agko-gw-1(config-router)#exit
msk-hostel-agko-gw-1(config)#^Z
msk-hostel-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-hostel-agko-gw-1#
```

Рисунок 4: Настройка маршрутизирующего коммутатора msk-hostel-agko-gw-1

```
sch-sochi-agko-gw-1>enable
Password:
sch-sochi-agko-gw-1#router ospf 1
                           ^
% Invalid input detected at '^' marker.

sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#router ospf 1
sch-sochi-agko-gw-1(config-router)#router-id 10.128.254.3
sch-sochi-agko-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-agko-gw-1(config-router)#exit
sch-sochi-agko-gw-1(config)#^Z
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-gw-1#
```

Рисунок 5: Настройка маршрутизатора sch-sochi-agko-gw-1

Теперь проверим состояние протокола OSPF на всех маршрутизаторах (рис. #fig:007 – #fig:009):

```
msk-q42-agko-gw-1>enable
Password:
msk-q42-agko-gw-1#sh ip ospf
 Routing Process "ospf 1" with ID 10.128.254.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 4
        Area has no authentication
        SPF algorithm executed 1 times
        Area ranges are
        Number of LSA 1. Checksum Sum 0x000bcc
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

msk-q42-agko-gw-1#sh ip ospf neighbor
```

```
msk-hostel-agko-gw-1>enable
Password:
msk-hostel-agko-gw-1#sh ip ospf
 Routing Process "ospf 1" with ID 10.128.254.4
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 2
        Area has no authentication
        SPF algorithm executed 1 times
        Area ranges are
        Number of LSA 1. Checksum Sum 0x00345a
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

msk-hostel-agko-gw-1#sh ip ospf neighbor

msk-hostel-agko-gw-1#sh ip ospf route
                                ^
% Invalid input detected at '^' marker.

msk-hostel-agko-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 2 subnets
C       10.129.1.0 is directly connected, Vlan202
C       10.129.128.0 is directly connected, Vlan301
S*   0.0.0.0/0 [1/0] via 10.129.1.1

msk-hostel-agko-gw-1#
msk-hostel-agko-gw-1#
```

```
sch-sochi-agko-gw-1#
03:33:21: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to FULL, Loading Done

03:33:21: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/0.7 from LOADING to FULL, Loading Done
sh ip ospf
 Routing Process "ospf 1" with ID 10.128.254.3
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 4
        Area has no authentication
        SPF algorithm executed 2 times
        Area ranges are
        Number of LSA 6. Checksum Sum 0x029677
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

sch-sochi-agko-gw-1#sh ip neighbor
                           ^
% Invalid input detected at '^' marker.

sch-sochi-agko-gw-1#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.128.254.1      1   FULL/DR         00:00:36    10.128.255.5    FastEthernet0/0.6
10.128.254.2      1   FULL/DR         00:00:36    10.128.255.9    FastEthernet0/0.7
sch-sochi-agko-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 17 subnets, 3 masks
O       10.128.0.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.1.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.3.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.4.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.5.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.6.0/24 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
O       10.128.255.0/30 [110/2] via 10.128.255.5, 00:00:12, FastEthernet0/0.6
```

Следующим шагом настроим линк 42-й квартал – Сочи (рис. #fig:010 – #fig:013):

```
provider-agko-sw-1>enable
Password:
provider-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-agko-sw-1(config)#vlan 7
provider-agko-sw-1(config-vlan)#name q42-sochi
provider-agko-sw-1(config-vlan)#exit
provider-agko-sw-1(config)#interface vlan7
provider-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up
no shutdown
provider-agko-sw-1(config-if)#exit
provider-agko-sw-1(config)#^Z
provider-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
```

```
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#interface f0/1.7
msk-q42-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up
encapsulation dot1Q 7
msk-q42-agko-gw-1(config-subif)#ip address 10.128.255.9 255.255.255.252
msk-q42-agko-gw-1(config-subif)#descriprtion sochi
                                        ^
% Invalid input detected at '^' marker.

msk-q42-agko-gw-1(config-subif)#description sochi
msk-q42-agko-gw-1(config-subif)#exit
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-q42-agko-gw-1#
```

Рисунок 10: Настройка маршрутизатора msk-q42-agko-gw-1

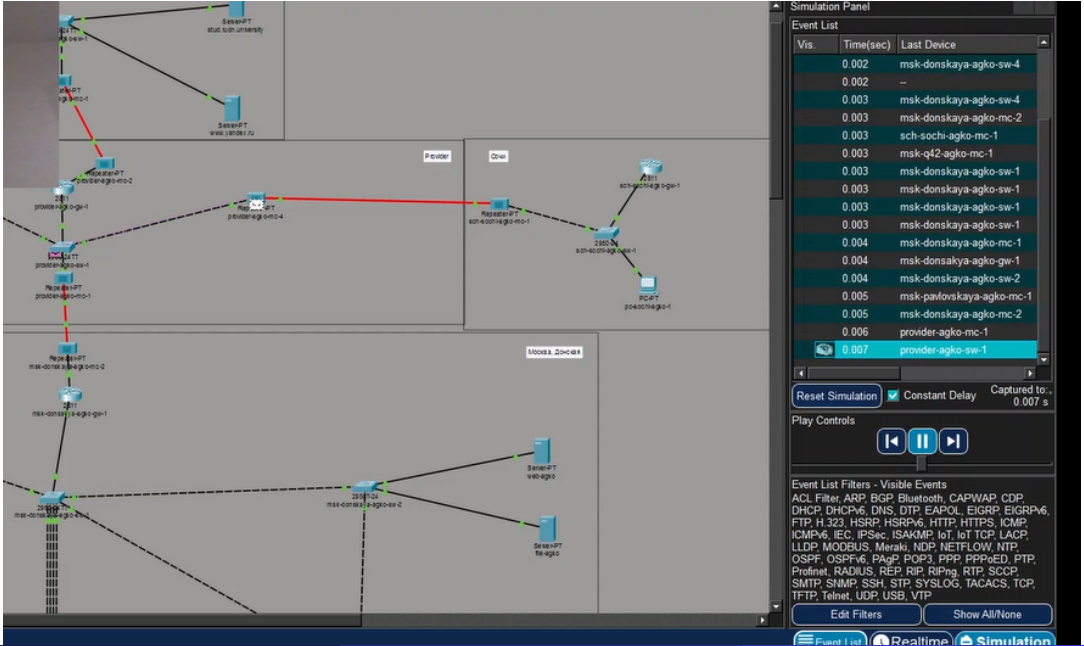
```
sch-sochi-agko-sw-1>enable
Password:
sch-sochi-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-sw-1(config)#vlan 7
sch-sochi-agko-sw-1(config-vlan)#name q42-sochi
sch-sochi-agko-sw-1(config-vlan)#exit
sch-sochi-agko-sw-1(config)#interface vlan7
sch-sochi-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up
no shutdown
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#^Z
sch-sochi-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-sw-1#
```

```
sch-sochi-agko-gw-1>enable
Password:
sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#interface f0/0.7
sch-sochi-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up
encapsulation dot1Q 7
sch-sochi-agko-gw-1(config-subif)#ip address 10.128.255.10 255.255.255.252
sch-sochi-agko-gw-1(config-subif)#description q42
sch-sochi-agko-gw-1(config-subif)#exit
sch-sochi-agko-gw-1(config)#^Z
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-gw-1#
```

Рисунок 12: Настройка маршрутизатора sch-sochi-gw-1

В режиме симуляции отследим движение пакета ICMP с ноутбука администратора сети на Донской в Москве (`admin-donskaya-agko`) до компьютера пользователя в филиале в г. Сочи (`pc-sochi-1`) (рис. #fig:014 – #fig:015):

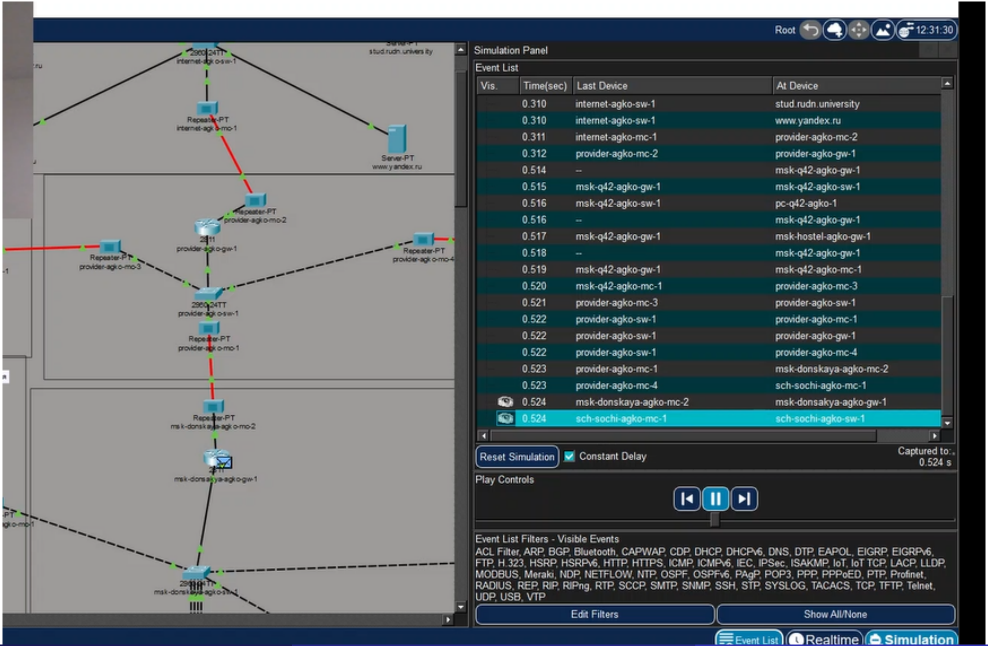
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```


Simulation Panel
Event List
Vis.
Time(sec)
Last Device
0.002 msk-donskaya-agko-sw-4
0.002 --
0.003 msk-donskaya-agko-sw-4
0.003 msk-donskaya-agko-mc-2
0.003 sch-sochi-agko-mc-1
0.003 msk-q42-agko-mc-1
0.003 msk-donskaya-agko-sw-1
0.003 msk-donskaya-agko-sw-1
0.003 msk-donskaya-agko-sw-1
0.003 msk-donskaya-agko-sw-1
0.004 msk-donskaya-agko-mc-1
0.004 msk-donsakya-agko-gw-1
0.004 msk-donskaya-agko-sw-2
0.005 msk-pavlovskaya-agko-mc-1
0.005 msk-donskaya-agko-mc-2
0.006 provider-agko-mc-1
0.007 provider-agko-sw-1
Reset Simulation
Constant Delay
Captured to:
0.007 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Realtime
Simulation

Следующим шагом на коммутаторе провайдера отключим временно vlan 6 и в режиме симуляции убедимся в изменении маршрута прохождения пакета ICMP (рис. #fig:016 – #fig:017):

```
Building configuration...
[OK]
provider-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-agko-sw-1(config)#no vlan 6
provider-agko-sw-1(config)#^Z
provider-agko-sw-1#
```

Рисунок 15: Временное отключение на коммутаторе провайдера vlan 6


Root
12:31:30
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device
0.310 internet-agko-sw-1 stud.rudn.university
0.310 internet-agko-sw-1 www.yandex.ru
0.311 internet-agko-mc-1 provider-agko-mc-2
0.312 provider-agko-mc-2 provider-agko-gw-1
0.514 -- msk-q42-agko-gw-1
0.515 msk-q42-agko-gw-1 msk-q42-agko-sw-1
0.516 msk-q42-agko-sw-1 pc-q42-agko-1
0.516 -- msk-q42-agko-gw-1
0.517 msk-q42-agko-gw-1 msk-hostel-agko-gw-1
0.518 -- msk-q42-agko-gw-1
0.519 msk-q42-agko-gw-1 msk-q42-agko-mc-1
0.520 msk-q42-agko-mc-1 provider-agko-mc-3
0.521 provider-agko-mc-3 provider-agko-sw-1
0.522 provider-agko-sw-1 provider-agko-mc-1
0.522 provider-agko-sw-1 provider-agko-gw-1
0.522 provider-agko-sw-1 provider-agko-mc-4
0.523 provider-agko-mc-1 msk-donskaya-agko-mc-2
0.523 provider-agko-mc-4 sch-sochi-agko-mc-1
0.524 msk-donskaya-agko-mc-2 msk-donsakya-agko-gw-1
0.524 sch-sochi-agko-mc-1 sch-sochi-agko-sw-1
Reset Simulation
Constant Delay
Captured to:
0.524 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Event List
Realtime
Simulation
Server-PT
www.yandex.ru
Repeater-PT
internet-agko-mc-1
Repeater-PT
provider-agko-mc-2
provider-agko-gw-1
Repeater-PT
provider-agko-mc-3
Repeater-PT
provider-agko-mc-4
2960-24TT
provider-agko-sw-1
Repeater-PT
provider-agko-mc-1
Repeater-PT
msk-donskaya-agko-mc-2
msk-donsakya-agko-gw-1

На коммутаторе провайдера восстановим vlan 6 и в режиме симуляции вновь убедимся в изменении маршрута прохождения пакета ICMP (рис. #fig:019):

```
provider-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-agko-sw-1(config)#vlan 6
provider-agko-sw-1(config-vlan)#name sochi
provider-agko-sw-1(config-vlan)#^Z
provider-agko-sw-1#wr m
Building configuration...
[OK]
provider-agko-sw-1#
```

Рисунок 17: Восстановление на коммутаторе провайдера vlan 6

Вывод

В ходе выполнения лабораторной работы мы настроили динамическую маршрутизацию OSPF между территориями организации (Донская, 42-й квартал, Сочи), проверили работу протокола в штатном режиме, а также при отключении и восстановлении линка, убедившись в автоматическом изменении маршрута.